OPENNESS OF VERSALITY FOR THE STACK OF CURVES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Remark quot-remark-curves-base-change (canonical tag 0D54). cited at statement line 4997. Base change remark.
- Equation quot-equation-curves-over-proper-spaces (canonical tag 0D50). cited at proof line 5003. The fully faithful embedding.
- Lemma quot-lemma-spaces-defo-thy (canonical tag 0D3X). cited at proof line 5005. Collected card: openness of versality for spaces.

Source reference. Original remark 0D54: sources/stacks-chow/tags.txt, source lines 16474–16474. Cited at statement line 4997: base change remark.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at statement line 5003: the fully faithful embedding.

Source reference. Original lemma 0D3X: sources/stacks-chow/tags.txt, source lines 16432–16432. Cited at statement line 5005: corresponding openness of versality for the stack of spaces.

Lemma 0.2. *The stack in groupoids $\mathcal{X} = \mathcal{C}urves$ satisfies openness of versality over* Spec($\mathbf{Z}$)*. Similarly, after base change (Remark 0D54) openness of versality holds over any Noetherian base scheme S.*

Proof. This is immediate from the fully faithful embedding (0D50) and the corresponding fact for $\mathcal{S}paces'_{fp,flat,proper}$ (Lemma 0D3X). □

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.